Interim Report

Analysis of electric vehicle charging station usage and profitability in the UK based on real-time data.

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Project and Dissertation in Data Science - CSC8639

1. Introduction:

Transportation is a major contributor to greenhouse gas emissions in the UK, which is becoming a factor in climate change, one of the biggest challenges of the 21st century.[10] Electric vehicles (EVs) represent a significant leap towards sustainable transportation, producing zero harmful emissions at the point of use. Some of the leading manufacturing companies of EVs in the UK include big names such as Tesla, MG, Kia, Volkswagen, etc. [1] However, widespread EV adoption depends on developing a robust and efficient charging infrastructure. Designing such an infrastructure presents a complicated challenge.[2] It needs to fulfil the diverse charging needs of EV owners, ensuring convenient access to charging points. At the same time, it must maintain high station utilization rates to be financially feasible. Most importantly, the infrastructure needs to minimize its overall environmental impact.[2][3][9]

While the focus often lies on the convenience and affordability of charging, understanding the relationship between charging patterns and carbon emissions is equally important.[10] Traditional gasoline-powered vehicles contribute significantly to greenhouse gas emissions, primarily through the burning of fossil fuels. While EVs themselves produce zero engine out emissions, the environmental impact of charging them depends on the source of the electricity used.[10] Understanding how charging patterns influence energy consumption, and the associated carbon footprint is essential for developing a truly sustainable charging network.[2][10]

This research project aims to leverage the power of data analytics to analyse and predict EV charging patterns in the UK. A detailed analysis of a comprehensive dataset will be carried out in this project which comprises data like energy consumption, EV arrival times, charger occupancy, utilization rates, and profitability of public charging stations in the UK.[2][3][4][9] By analysing these factors, various key trends will be identified, and significant insights will be drawn.[4] Also, the associated carbon emissions data will be analysed to understand the effect of variable energy consumption on carbon emissions.[10] All these detailed analyses will contribute to developing empirical models. These models will serve as valuable tools for researchers, policymakers, and industry stakeholders. They will be able to:

- Simulate and test various charging infrastructure scenarios that consider both environmental and economic factors. This will allow them to identify the most efficient deployment strategies for charging stations, maximizing convenience and profitability while minimizing carbon footprint.
- Verify existing models related to EV charging behaviour and their impact on the environment. This will contribute to a more comprehensive understanding of how charging patterns influence overall carbon emissions.
- Make informed decisions to optimize charging infrastructure usage, profitability, and environmental impact. This will ultimately pave the way for a more sustainable transportation network in the UK.

By analysing real-time EV charging data and incorporating carbon emissions data, this research project aims to bridge the gap between convenience, economic viability, and environmental responsibility in developing an optimal EV charging infrastructure for the UK.

2. Aims & Objectives:

2.1 Aim

To analyse the usage patterns and profitability of electric vehicle charging stations in the UK using real-time data, with a focus on optimizing the infrastructure for both economic feasibility and environmental sustainability.

2.2 Objectives

- Collect and preprocess real-time data from various public EV charging stations across the UK.
- Identify and analyse key factors influencing the utilization rate of EV charging stations which include the energy consumption, charging durations, charger occupancy and peak hours.

- Associate the carbon intensity data to understand the relationship between energy consumption and carbon emissions over time.
- Estimate the potential carbon emission reduction achievable through optimized charging infrastructure deployment.
- Extract significant insights and suggest recommendations for optimizing charging infrastructure usage, profitability, and environmental impact in the UK.
- Develop models to predict future test scenarios of energy consumption.

3. Overview of progress:

3.1 Literature Review:

A detailed review of existing research on EV charging infrastructure and profitability analysis was carried out. This review has identified key factors that influence charging station usage and financial performance.[2][3][9] Key insights from the literature include the importance of station location, pricing strategies, and the integration of renewable energy sources.[2][3][4][9][10]

3.2 Tools and Techniques:

Python has been the primary tool for data analysis, utilizing libraries such as Glob for data extraction, Pandas for data manipulation and Plotly for visualization. These tools have enabled efficient handling and analysis of large datasets.

3.3 Data Collection:

Data collection efforts have focused on gathering real-time data from public EV charging stations across the UK. The data was publicly accessible and was collected from sources like "London Open Barnet Data – Barnet Council" and "ChargePlace Scotland".[5][7][8] The data consists of energy consumption, EV arrival times, charger occupancy and utilization rates. Additionally, carbon emissions data has been sourced from "Carbon Intensity API" to assess the environmental impact of different charging patterns.[11]

3.4 Data Processing:

Initial steps have been taken to clean and preprocess the collected data. This includes handling missing values, normalizing data formats, and ensuring data consistency. Preliminary exploratory data analysis (EDA) has been performed to identify any obvious trends and anomalies in the data. Figure 1 shows the frequency distribution of Energy Consumption of London Barnet Data. The peak energy consumption appears between a range of 7.5 to 12.5 kWh. Furthermore, the carbon intensity data was integrated with the energy consumption data and the carbon emissions for each event were calculated as [Carbon Emissions (gCO_2) = Carbon Intensity (gCO_2/kWh) * Energy Consumption (kWh)].

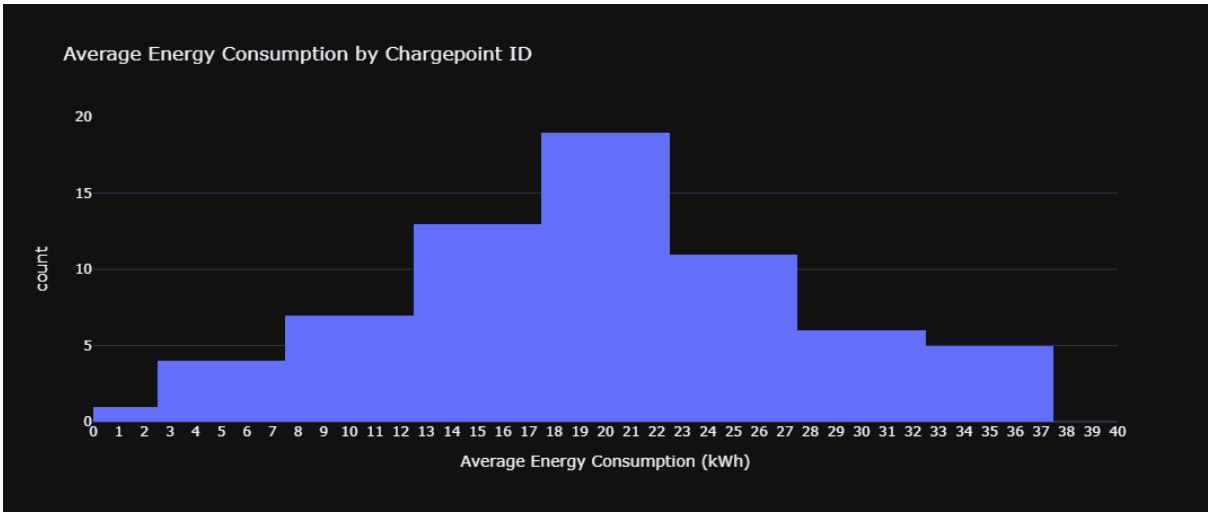

Figure 1 Distribution of Energy Consumption (kWh)

3.5 Exploratory Data Analysis (EDA):

Early EDA results have revealed several interesting patterns. For example, peak usage times have been identified, highlighting periods of high demand. The charging events were grouped according to their charging IDs and further grouped based on energy consumption ranges as shown in Figure 2 (Group 1 – 0 to 10.99 kWh, Group 2 – 11 to 20.99 kWh, Group 3 – 21 to 30.99 kWh, Group 4 – 31 to 40 kWh). Variations in charger occupancy have also been noted. Next, the integrated carbon intensity data shows a relation between energy consumption and carbon emissions over time in Figure 3.

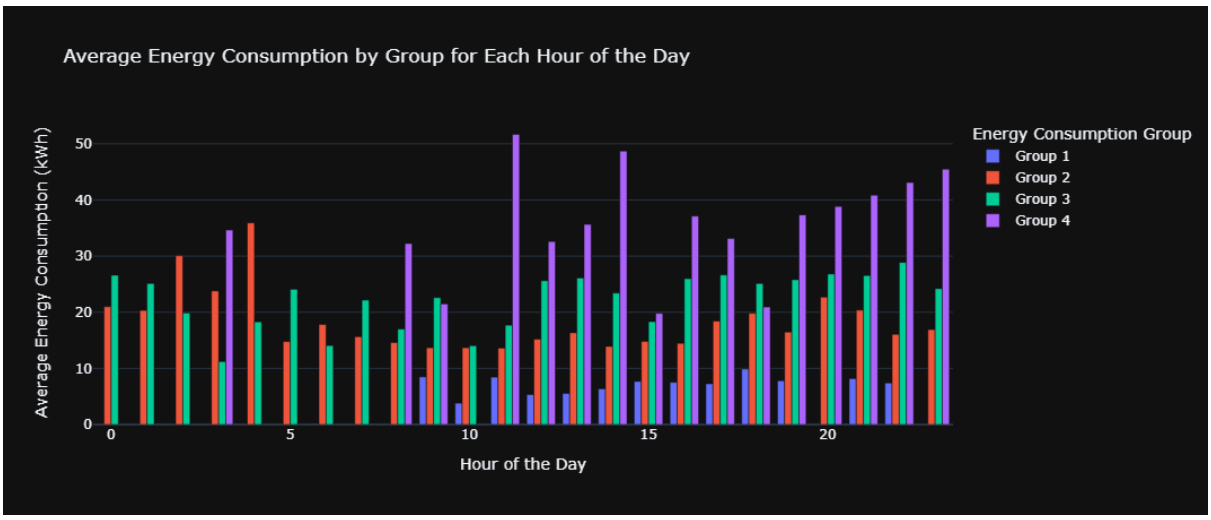

Figure 2. Average Energy Consumption by Group for Each hour of the day

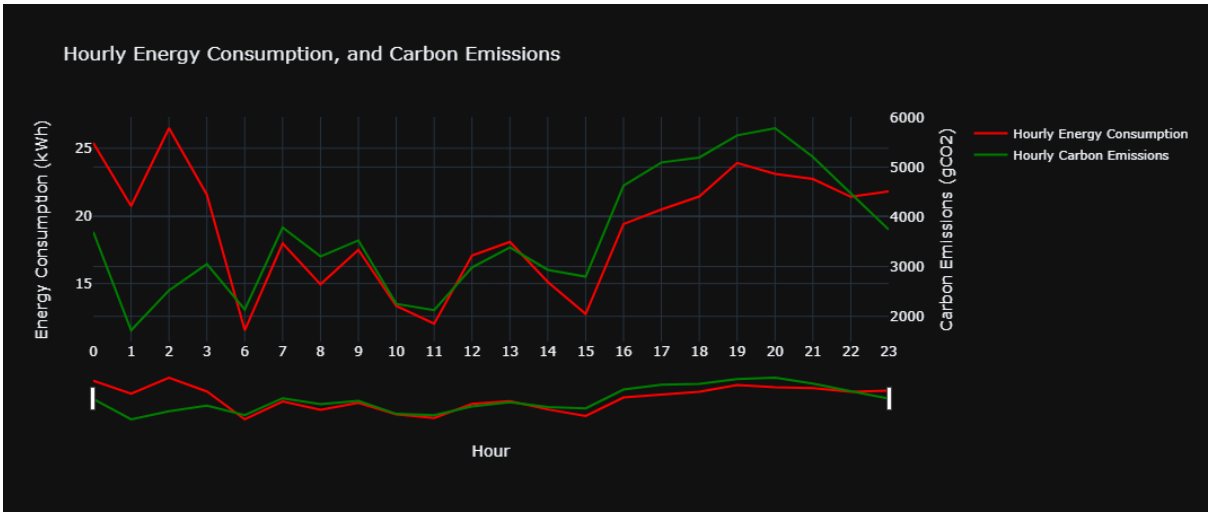

Figure 3 Hourly Energy Consumption and Carbon Emissions

3.6 Challenges Faced:

A few challenges have been encountered, including inconsistencies in data formats and incomplete data from some sources. These issues have been addressed through rigorous data cleaning procedures and by working closely with data providers to fill gaps. Another challenge has been the integration of carbon emissions data with charging data, which requires careful alignment of datasets based on time and location.

3.7 Next Steps

Moving forward, the focus will be on conducting detailed statistical and machine learning analyses to uncover deeper insights into usage patterns and profitability. This will include the development of empirical models to predict future usage and assess the impact of different factors on station performance. The integration of

carbon emissions data will also be refined to provide a comprehensive view of the environmental impact of EV charging in the UK.

4 Project Plan:

The project time frame spans from 1st May 2024 to 15th August 2024. The detailed plan is provided in Figure 4 in the form of a Gannt Chart. Initial work included an extensive literature review of recent electric charging trends and how to examine and identify the key features contributing to the infrastructure. The implementation of this project started on 16th May 2024 with Data Collection and understanding of the data. Further analysis of different regions of the UK will be done and integrated into the project. The carbon emissions for those regions will also be analysed. After successful completion of EDA, machine learning model development will begin to predict the future impact, usage and profitability. All significant insights drawn from the analysis will be evaluated by the stakeholders. Finally, all the findings will be compiled into a comprehensive report and will be reviewed to make necessary revisions based on the feedback.

5 References:

[1] Tom Swallow - Top EV manufacturing companies in the UK. *Retrieved from https://evmagazine.com/top10/top-10-best-selling-evs-in-the-uk - EV Magazine. (2023.)*

[2] Hecht, Christopher et al. - Analysis of electric vehicle charging station usage and profitability in Germany based on empirical data - *iScience, Volume 25, Issue 12, 105634 (2022)*

[3] Christopher Hecht, Saurav Das, Christian Bussar, Dirk Uwe Sauer - Representative, empirical, real-world charging station usage characteristics and data in *Germany, eTransportation, Volume 6, 2020, 100079, ISSN 2590-1168, https://doi.org/10.1016/j.etran.2020.100079. (https://www.sciencedirect.com/science/article/pii/S2590116820300369)*

[4] Jade Edwards - EV charging statistics 2024. *Retrieved from https://www.zap-map.com/ev-stats/how-many-charging-points – Zapmap (2024)*

[5] London Borough of Barnet - Electric Vehicle Charging Transactions. *Retrieved from https://www.data.gov.uk/dataset/16c7326b-57fe-4803-88f8-9286c387f68a/electric-vehicle-charging-transactions – data.gov.uk | Find open data (2024)*

[6] SP Energy Networks partnered with PTV Group, Smarter Grid Solutions and EA Technology - The Charge Project. - *Retrieved from SP Energy Networks (2022)*

[7] ChargePlace Scotland - Monthly Charge Point Performance. *Retrieved from https://chargeplacescotland.org/monthly-charge-point-performance/ - ChargePlace Scotland (2024)*

[8] Barnet Open Data - Electric Vehicle Charging Transactions. *Retrieved from https://open.barnet.gov.uk/dataset/exm5m/electric-vehicle-charging-transactions – Barnet London Borough (2022)*

[9] Safak Bayram, Ali Saad, Ryan Sims, Colin Herron, Stuart Galloway - Usage Analysis of Public AC Chargers in the UK - *Department of Electronic and Electrical Engineering, University of Strathclyde, Glasgow, UK. Connected Kerb, 51- 52 Frith Street, London, UK. Power Networks Demonstration Centre, University of Strathclyde, Glasgow UK. School of Engineering, Newcastle University, Newcastle, UK (2023)*

[10] Daniel Mehlig, Helen ApSimon, Iain Staffell -Emissions from charging electric vehicles in the UK - *Transportation Research Part D: Transport and Environment, Volume 110, 2022, 103430, ISSN 1361-9209, https://doi.org/10.1016/j.trd.2022.103430. (https://www.sciencedirect.com/science/article/pii/S1361920922002565)*

[11] National Grid ESO, Environmental Defense Fund Europe, University of Oxford Department of Computer Science and WWF - Carbon Intensity Data – *(https://carbonintensity.org.uk/) Carbon Intensity API, National Grid ESO.*

Gannt Chart:

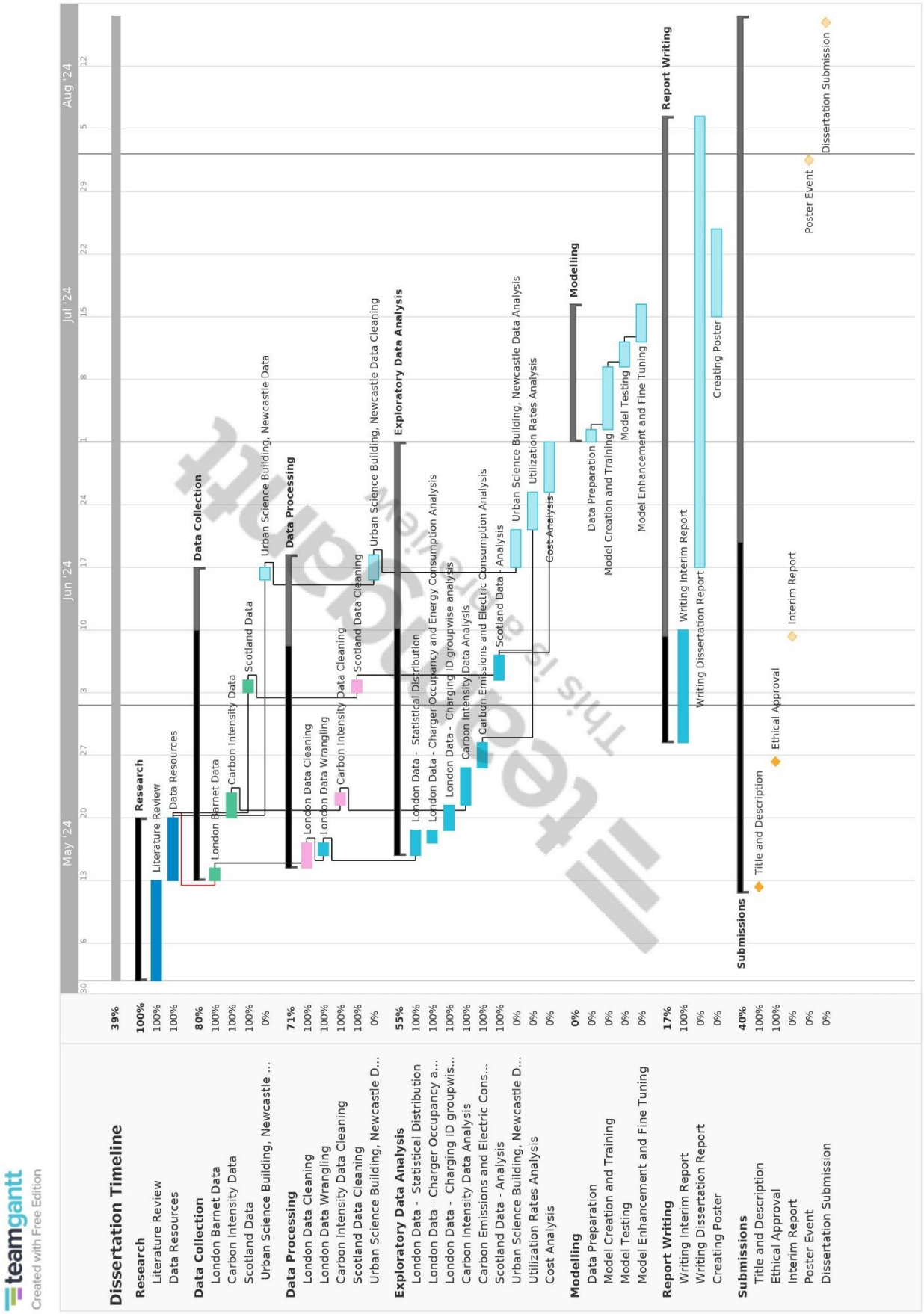

Figure 4 Gannt Chart representing the project plan. The tasks are to scale as per date range and dependencies are shown as lines

6 Data Management Plan:

0. Project Title, author, version and date	
Project Name: - *Analysis of electric vehicle charging station usage and profitability in the UK based on real-time data.*	
Author: *Kaustubh Kulkarni*	***Date:*** *07th June 2024*

1. Description of the data

1.1 Type of study

This study involves analyzing real-time data from public EV charging stations to understand usage patterns, energy consumption, and profitability, as well as assessing the environmental impact through carbon emissions data.

1.2 Types of data

Quantitative data generated from electronic records of EV charging stations, including energy consumption, EV arrival times, charger occupancy, utilization rates, and profitability metrics. Additionally, carbon emissions data will be included to analyze the environmental impact.

1.3 Format and scale of the data

Data will be stored in CSV format and SQL databases, analyzed using Python and R. The dataset will include thousands of records detailing various aspects of EV charging station usage and profitability, covering multiple locations and periods.

2. Data collection/generation

2.1 Methodologies for data collection/generation

Data will be collected from government websites with records of public EV charging stations through automated logging systems. [London Barnet Data, Scotland data] Additional data sources include carbon intensity data for carbon emissions associated with energy consumptions. Standard data collection methods are used to ensure consistency and accuracy. Data was downloaded directly from the available websites, although the carbon intensity data was downloaded every single day and then merged with the EV data for analysis.

2.2 Data quality and standards

The data quality was assessed based on the consistency and availability of the data. The data was easily available on public sites. The EV data had many inconsistencies like the occupancies and consumption times were not in proper datetime format. There were many null and duplicate values. All these issues were taken care of by Data Cleaning and Data Wrangling processes.

3. Data management, documentation and curation

3.1 Managing, storing and curating data.

Data will be downloaded to the local system and will be stored in the project folder uploaded to the cloud (OneDrive). Hence, the data along with the project will be easily accessible. Further, the project is being backed up in the GitHub repository after every update. Data management will adhere to FAIR (Findable, Accessible, Interoperable, Reusable) principles.

3.2 Metadata standards and data documentation

Detailed metadata will include descriptions of data collection methods, data sources, variable definitions, and coding schemes. Metadata will be stored in a ReadME.md file in the Git repository which can be referred and will be easily understandable.

3.3 Data preservation strategy and standards

Data will be preserved in a secure, long-term repository with infinite retention period.

4. Data security and confidentiality of potentially disclosive information

4.1 Formal information/data security standards

This research does not involve any human participant or any personal data. Hence, there is no risk of security.

4.2 Main risks to data security

Since data is entirely public data, security issues with third party storage solutions (raised by ncl.ac.uk https://www.ncl.ac.uk/library/academics-and-researchers/research/rdm/working/) are not a concern. The project is uploaded on the OneDrive storage provided by the university.

5. Data sharing and access

5.1 Suitability for sharing

Yes. The data used in my project is publicly available on referenced websites. Hence, the data used in the study is suitable for sharing as it provides valuable insights into EV charging patterns, which can benefit researchers, policymakers, and industry stakeholders.

5.2 Discovery by potential users of the research data

The research data and summary information will be made available through the Git repository to be discovered by potential users. If the project is successful in fulfilling its aim, then a research paper on the same will be published and a DOI will be generated to facilitate discovery.

5.3 Governance of access

Since this research is not population health and patient-based research, this section is not applicable.

5.4 The study team's exclusive use of the data

Data is stored in the Git Repository along with the project code for a long timeline and data management will adhere to FAIR (Findable, Accessible, Interoperable, Reusable) principles. This will make the data to be easily accessible and usable.

5.5 Restrictions or delays to sharing, with planned actions to limit such restrictions

The data sharing procedures will be clearly outlined in the repository's README file. Since the datasets utilized are fully public, there are no confidentiality concerns. Any intellectual property derived from the developed models will be openly shared to foster progress in the field, with terms of use clearly detailed in the project README.

5.6 Regulation of responsibilities of users

Since this research is not medical based research, this section is not applicable.

6. Responsibilities and Resources

Any resources required to fulfil the plan? • *PyCharm Student Edition* • *Public Data Access* • *OneDrive Storage*
7. Relevant institutional, departmental or study policies on data sharing and data security
Please complete, where such policies are (i) relevant to your study, and (ii) are in the public domain, e.g. accessible through the internet.

Policy	**URL or Reference**
Data Management Policy & Procedures	*https://www.go-fair.org/fair-principles/*
Data Security Policy	*https://www.ncl.ac.uk/library/academics-and-researchers/research/rdm/working/*

8. Author of this Data Management Plan (Name) and, if different to that of the Principal Investigator, their **telephone & email contact details**
Kaustubh Kulkarni